

Power Transmission

→ Transmission Systems (or) Drives :-

- The Rotational motion can be transmitted from one mechanical element to the other with the help of certain systems which is known as Transmission Systems (or) drives

→ Transmission Systems are classified depending on,

- | | | |
|---------------------------|---|------------------|
| (i) distance between them | } | (1) Belt drives |
| (ii) Speed | | (2) Rope drives |
| (iii) Power | | (3) Chain drives |
| | | (4) Gear drives |

[*] Belt drives

Belt :- • A belt is a thin inextensible band

- Power Transmission belts are generally made from leather, rubber, canvas and balata.

[balata tree] E.g.: golf balls (Red timber)

Belt drives :-

- Belt drives are used when power has to be transmitted between two shafts.
- It consists of two pulleys which is mounted on two different shafts. i.e. driving shaft and driven shaft.

The driving shaft receives the power from the main and it transmits power to the driven shaft.

The pulley mounted on the driving shaft is called driving pulley (or) driver.

- The pulley mounted on the driver shaft is called as driver pulley (or) follower.
- An endless belt passes over these pulleys and is kept in tension to avoid slip over the pulley.
- The portion of belt which is having less tension is called Slack Side [Loose side]
- The portion of belt which is having higher tension is called Tight Side.
- The Rotation of the driver pulley is caused due to the difference in tensions on the Slack Side & Tight Sides.
- The portions of the belt between the two pulleys that become either Slack (or) tight depends on the direction of Rotation of the driving pulley.
- For clockwise Rotation of the driving pulley, The lower Side of the belt will be the tight Side and the upper Side will be the Slack Side.
- The tension in the tight and Slack Sides of the belt depends on the angle of contact, the belt drives have to be arranged such that the Slack Side comes above and the tight Side comes below the pulleys.
- This arrangement increases the angle of contact of the belt on the driver Side.
- Sometimes in a belt drive there is always a possibility of some slipping takes place between the belt and the pulleys which causes the driver pulley to rotate at a lesser speed, consequently reduces the power transmission.
- Hence belt drives are said to be not a positive type of power transmission system.

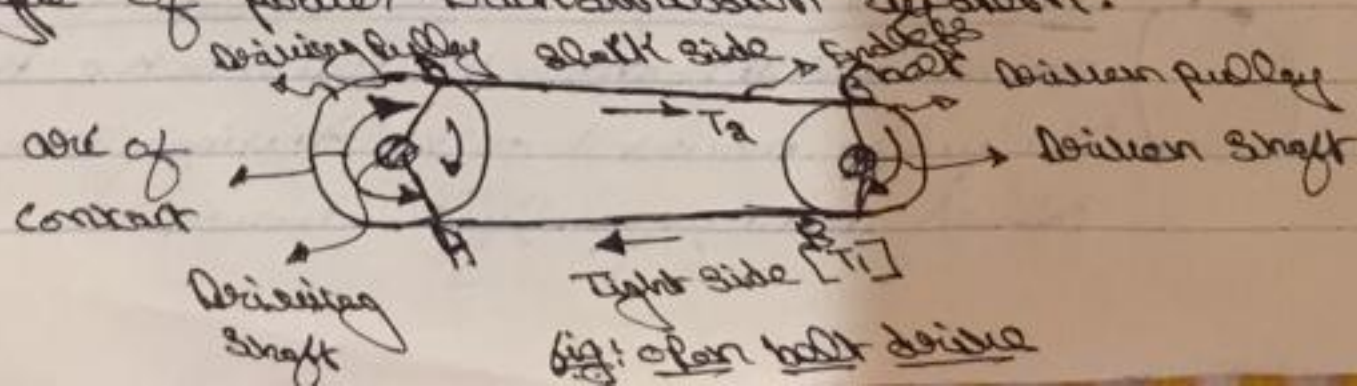


Fig: open belt drive

[*] 2 Types of Belt drives ~

The two types of Belt drives are

- ① open belt drive
- ② crossed belt drive.

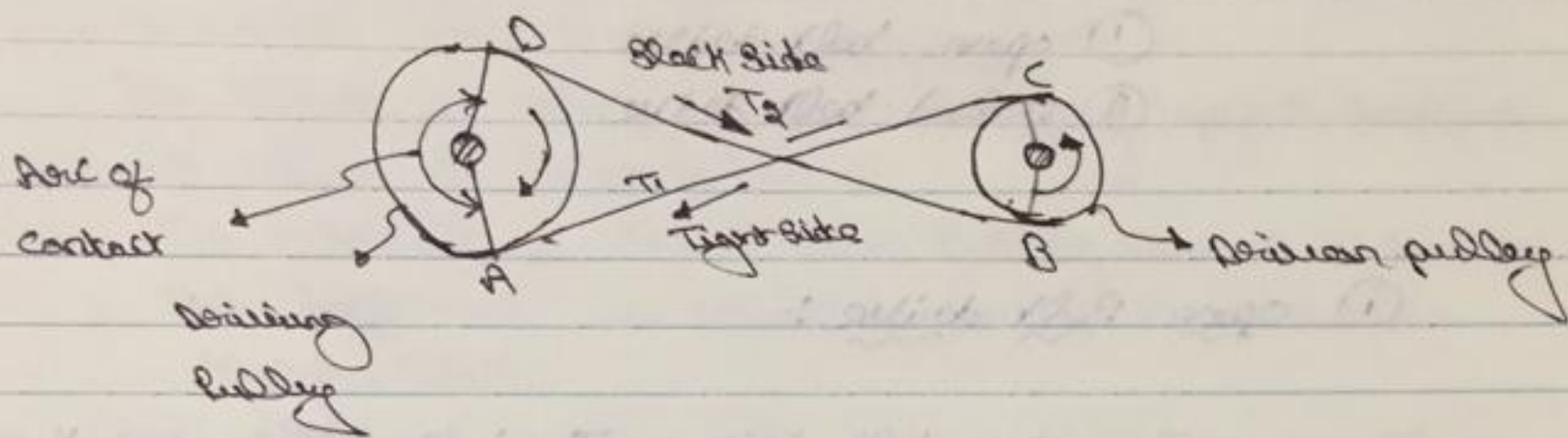
① open Belt drive :-

- In open belt drive, The belt passes over the driver and the driven pulleys in an open loop.
- Thus, the Shafts Rotate in the same direction.
- The driver pulls the belt from lower side ~~BA~~ and delivers it to upper side ~~DC~~.
- Since there is more tension in the lower side, the lower side of the belt should be the tight side & upper side should be the slack (or) loose side.

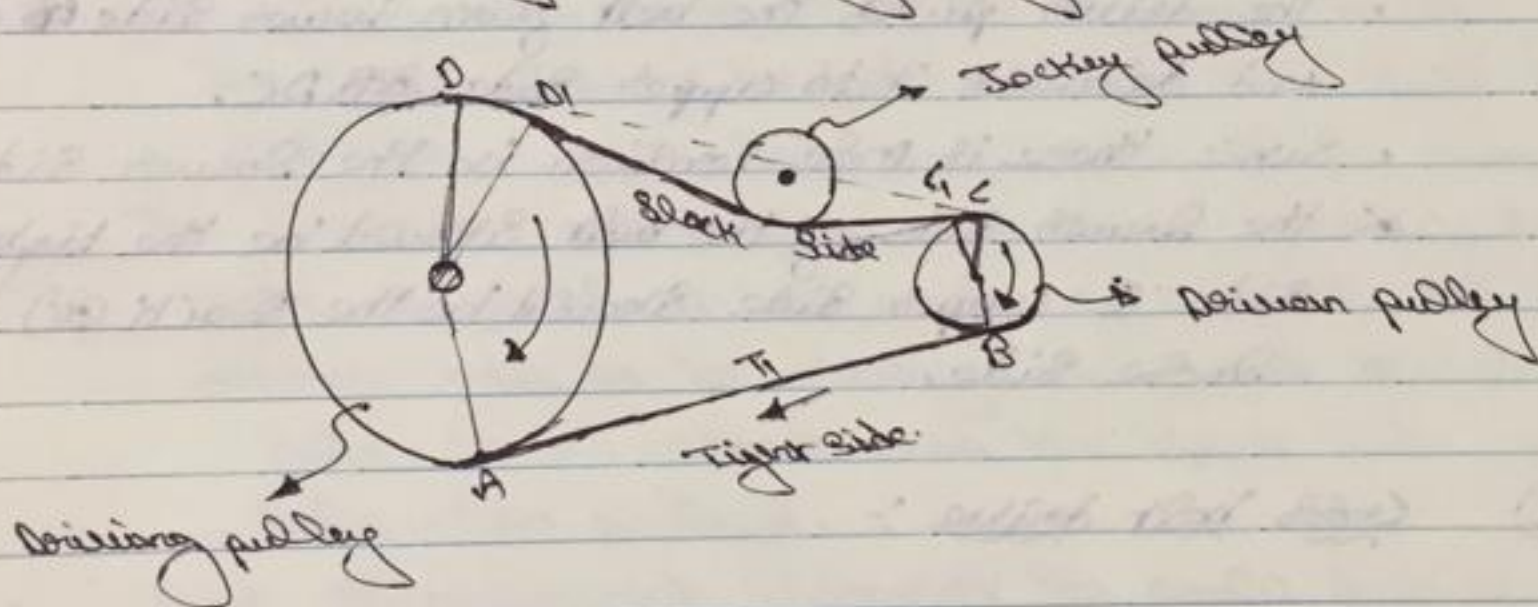
② Crossed belt drive :-

- In crossed belt drive, The belt passes over the driver pulleys and the driven pulleys in a crossed loop.
- Thus the Shafts Rotate in the opposite directions.
- At the Junction where the belt crosses, it rubs against itself and wears off.
- To avoid excessive wear, the Shafts must be placed at a maximum distance from each other and operated at very low speeds.

- The driver pulls the belt from one side CA [tight side] and delivers it to another side DB [slack side]

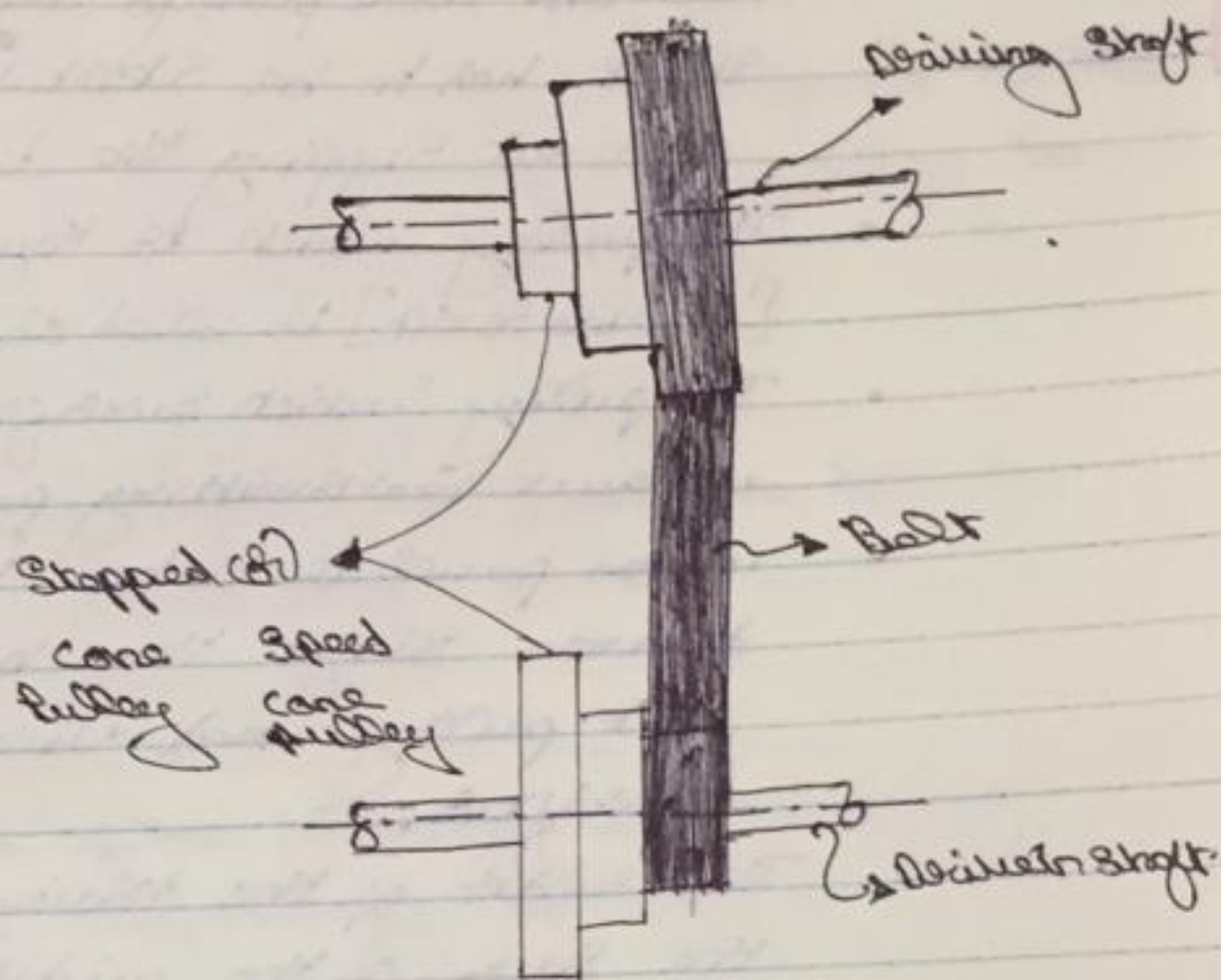


[*] Idler pulley (or) Jockey pulley



- In an open belt drive arrangement, if the centre distance is small, (or) if driven pulley is very small, then the arc of contact of the belt with the driven pulley will be very small, which reduces the tension in the belt, (or) if the required tension of the belt cannot be obtained.
- To obtain the required tension of the belt, Idler (or) Jockey pulley is placed on the slack side of the belt as shown in fig.
- This increases the arc of contact and the tension of the belt which results in increased power transmission.

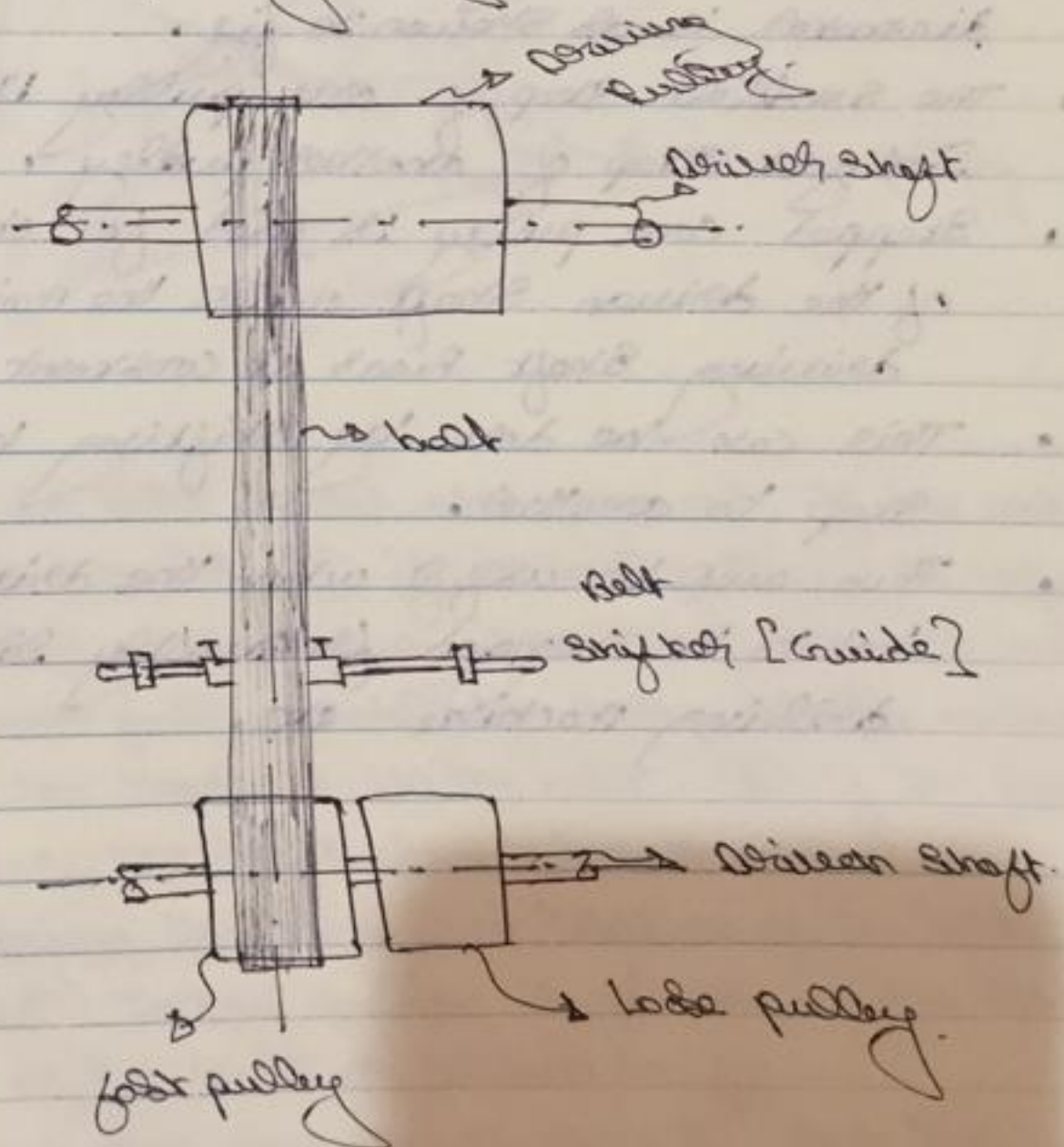
[*] Stepped cone pulley (or) Speed cone pulley :-



- Stepped cone pulley has several steps of varying diameter as shown in fig.
- The smallest step of one pulley is opposite to the largest step of another pulley.
- Stepped cone pulley is used for changing the speed of the driven shaft while the main shaft (or) driving shaft runs at constant speed.
- This can be done by shifting belts from one step to another.
- This will be useful when the driven shaft speed has to be changed frequently like lathes, drilling machine etc.

* Fast and loose pulley:-

- Fast and loose pulleys are used when the driven shaft has to be started (~~stopped~~) stopped when desired, without stopping the driving shaft.
- The pulley which is keyed to the machine shaft [driven shaft] is called as fast (~~or~~) tight pulley.
- The pulley which runs freely on the driven shaft without transmitting power is called loose pulley.
- When power is to be transmitted to the driven shaft then belt is pushed to the fast pulley and vice versa using a belt shifter.
- The width of the driving pulley is equal to the sum of the widths of the fast and loose pulley to gether.



* Velocity Ratio of Belt Drive

Velocity Ratio of a belt drive is defined as the Ratio of the Speed of the driving pulley to the Speed of the driven pulley.

Let d_1 and d_2

$$\text{Velocity Ratio} = \frac{N_1}{N_2}$$

Diagoneters of the driving and the driven pulley respectively, in m.

N_1 & N_2 :-

Speed of the driving and the driven pulley respectively, in RPM.

* $\rightarrow$ Assuming that there is no slip between the belt and pulley rim, the linear speed at every point on the belt must be same.

$$\pi d_1 N_1 = \pi d_2 N_2$$

$$d_1 N_1 = d_2 N_2$$

$$\text{Velocity Ratio} = \frac{N_1}{N_2} = \frac{d_2}{d_1}$$

The circumferential Speed of driving and driven pulleys and the linear Speed of the belt are equal

* Creep:-

- In a belt drive, The straight portions of belt will be alternately subjected to high & low tensions.
- The tight side which is subjected to high tension is subjected to stretching.
- The slack side, which is subjected to low tension is subjected to contraction.
- Since the stretching is more than contraction there will be an increased length of belt which results in a relative motion between belt & pulley surface.
- This relative motion is called Creep.
- Creep results in the loss of power transmission & decrease in the velocity ratio.

* Slip:-

- In a belt drive when the difference b/w the tensions in the tight & slack sides of the belt $[T_1 - T_2]$ is equal to the force of friction, The friction between pulley and belt will provide the necessary frictional grip to avoid the sliding of the belt over the pulley.
- If $[T_1 - T_2]$ is greater than the force of friction then the belt begins to slide over the surface of the pulley.
- This sliding of the belt which causes a relative motion between pulley and belt is called slip.
- Slip mainly occurs when $[T_1 - T_2]$ is very large (or) when the co-efficient of friction b/w the belt & pulley is decreased owing to the stretch of the belt (or) when the smoothness of the pulley surface is poor.

$$\text{Velocity Ratio} = \frac{v_1}{v_2} = \frac{d_2}{d_1} \left(\frac{100}{100 - s} \right) //$$

$$s = \frac{v_1 - v_2}{v_1} \times 100$$

Gear Drives* Gear drives:→ Gear:

- A gear is a toothed wheel
- The number of teeth cut on the periphery of a cylinder (or) cone (or) elliptical discs.
- gear is mounted on the shafts and keyed to them

→ Gear drives:

- Gear drives find a very prominent place in mechanical power transmission.
- Gear drives are preferred when considerable power has to be transmitted from one mechanical component to other mechanical component over a short centre distance with constant velocity ratio.
- Gear drives which consist of two gears, one is driving gear and another is driven gear.
- Driving gear mounted on driving shaft.
or Driven gear mounted on driven shaft.
- The gears are arranged so that the teeth of one will mesh with the teeth of the other.
- Hence the turning of one gear will cause turning of the other in the opposite direction.

[*] If the two gears have the same number of teeth → Both the magnitude & speeds are same.

[*] If the two gears have different number of teeth → Both the magnitude & speeds are changed.

[*] Let, a small gear [less teeth] that drives a larger gear [more teeth] $\rightarrow$ This arrangement will Increase the Torque and Reduce the Speed of the driven gear.

[*] Let, a larger gear [more teeth] that drives a smaller gear [less teeth] $\rightarrow$ This arrangement will decrease the Torque and Increase the Speed of the driven gear.

* Types of Gears :-

- The gears are generally classified as per the position of axes of the two shafts b/w which the motion is to be transmitted.

- ① parallel shaft axis gears.
- ② Intersecting shaft axis gears.
- ③ non intersecting & perpendicular shaft axis gears.
- ④ non intersecting and non-perpendicular shaft axis gears.

① Parallel shaft axis gears :-

- ② $\rightarrow$ Spur gears [*]
- $\rightarrow$ Helical gears [*]
- $\rightarrow$ Rack and Pinion [*]
- $\rightarrow$ Internal gears
- $\rightarrow$ Herringbone gears

② Intersecting shaft axes gears

→ Bell gears. [*]

③ non intersecting and perpendicular shaft gears

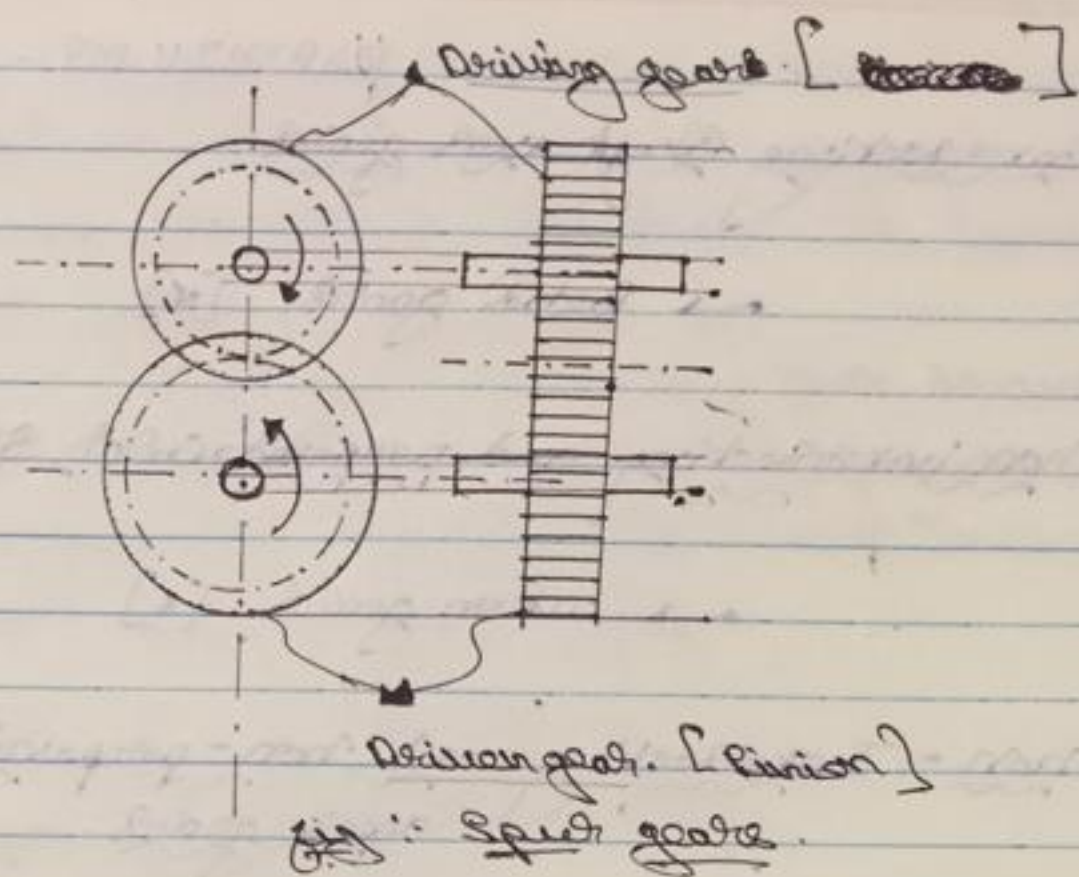
→ worm gears. [*]

④ non - Intersecting and non - perpendicular shaft axes gears

→ Spiral gears

[*] Spiral gear:

- Spiral gears are used to transmit motion between two parallel axes shaft.
- They are the simplest of all the gears and easiest in production.
- The teeth of spiral gears are cut along the periphery and parallel to the axis of gear.
- The contact between the mating gears will be along a line, hence spiral gears can transmit higher power.
- Spiral gears can be made of steel, brass, other metals and plastics.
- They are widely used in machine tools, automobile gear boxes and in all general cases of power transmission where gear drives are preferred. watches etc.



Advantages:-

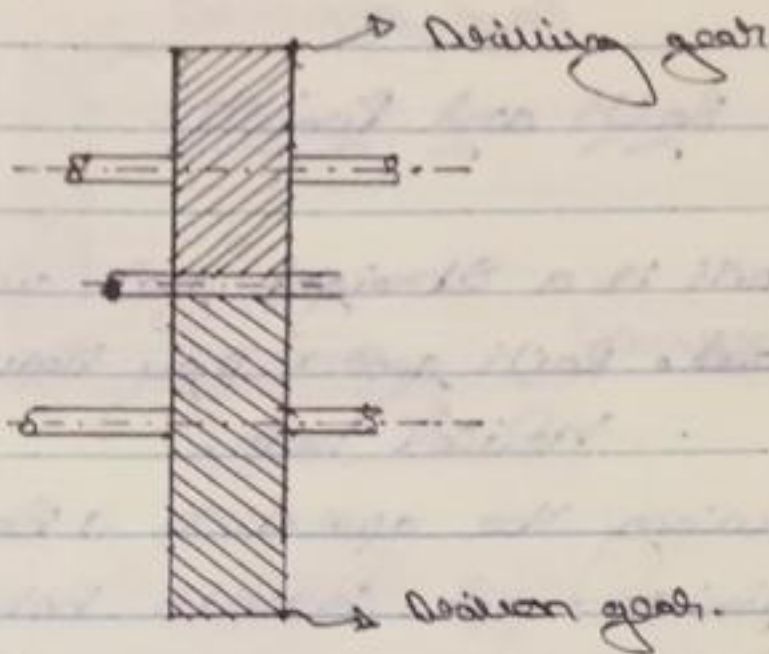
- Spur gears are easy to manufacture.
- They can make in variety of sizes from less than 25mm to several cm in dia.
- These gears are less expensive.
- Efficiency of Spur gear is high [upto 98%].

Disadvantages:-

- Spur gears are not used for high speed applications.
- The operation of Spur gear is noisy.

[*] Helical gears:-

- Helical gears are similar to spur gears but its tooth are cut at an angle with the axis of rotation of the gear.
- Helical gears can transmit motion from one shaft to another shaft which are parallel to each other.
- The helical angles of driving gear & driven gear are same in magnitude but of different hands.



Advantages:-

- Helical gears operate more quietly and smoothly than the spur gears.
- Teeth of helical gears are stronger than the spur gear teeth.
- Efficiency of helical gears is high [up to 99%].
- These gears are suitable for high speed applications.

Disadvantages

- Helical gears are difficult to manufacture as compared to spur gears.
- In helical gears, the friction, heat generation & wear is high.
- These gears are more expensive than the spur gears.

Applications:-

- Helical gears are commonly used in high speed automobile gear box, machine tool gear box, rolling mills, steam and gas turbines and other high speed applications.

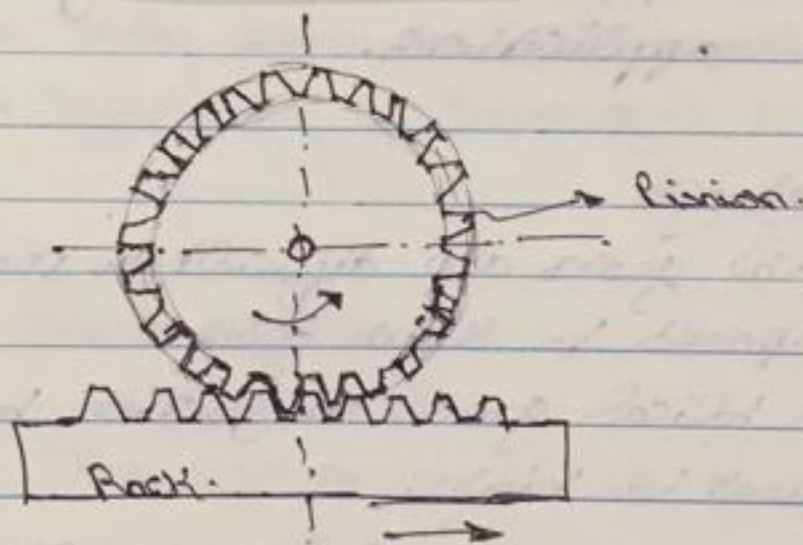
[*]

Rack and Pinion:-

- Rack is a Straight gear which has no curvature ~~and~~. Rack gears may have either Straight (or) helical teeth.
- During the operation, a Rack gear meshes with a pinion and converts Rotary motion of pinion into Reciprocating motion of Rack (or) Reverse.

Applications:-

- Rack and Pinion arrangement is used in feeding mechanisms, Reciprocating drives etc.
- They are also used in gauges and small equipments and lathes, milling, drilling and other machine tools.



[*]

Bevel gears →

- Bevel gears are commonly used for transmitting the motion between two intersecting shafts.
- In bevel gears teeth are cut on the tapered surface [conical surface].
- If the intersecting angle is exactly 90° then it is called as miter gear, if it is less than 90° then it is called as crown gear.
- Bevel gears are available with straight and helical teeth.

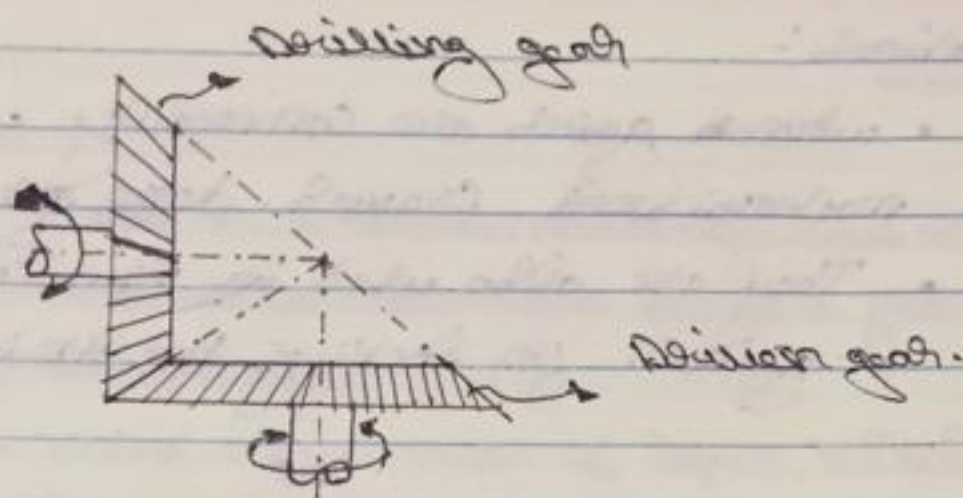


fig: Bevel gear

[*] worm and worm wheel:

- worm and worm gear are used for transmitting the motion between to non-intersecting and perpendicular shafts.
- It consists of simple worm (screw) and simple helical gear.
- worm and worm gears are used when large speed reduction ratio is required from one shaft to other shaft.

Advantages:-

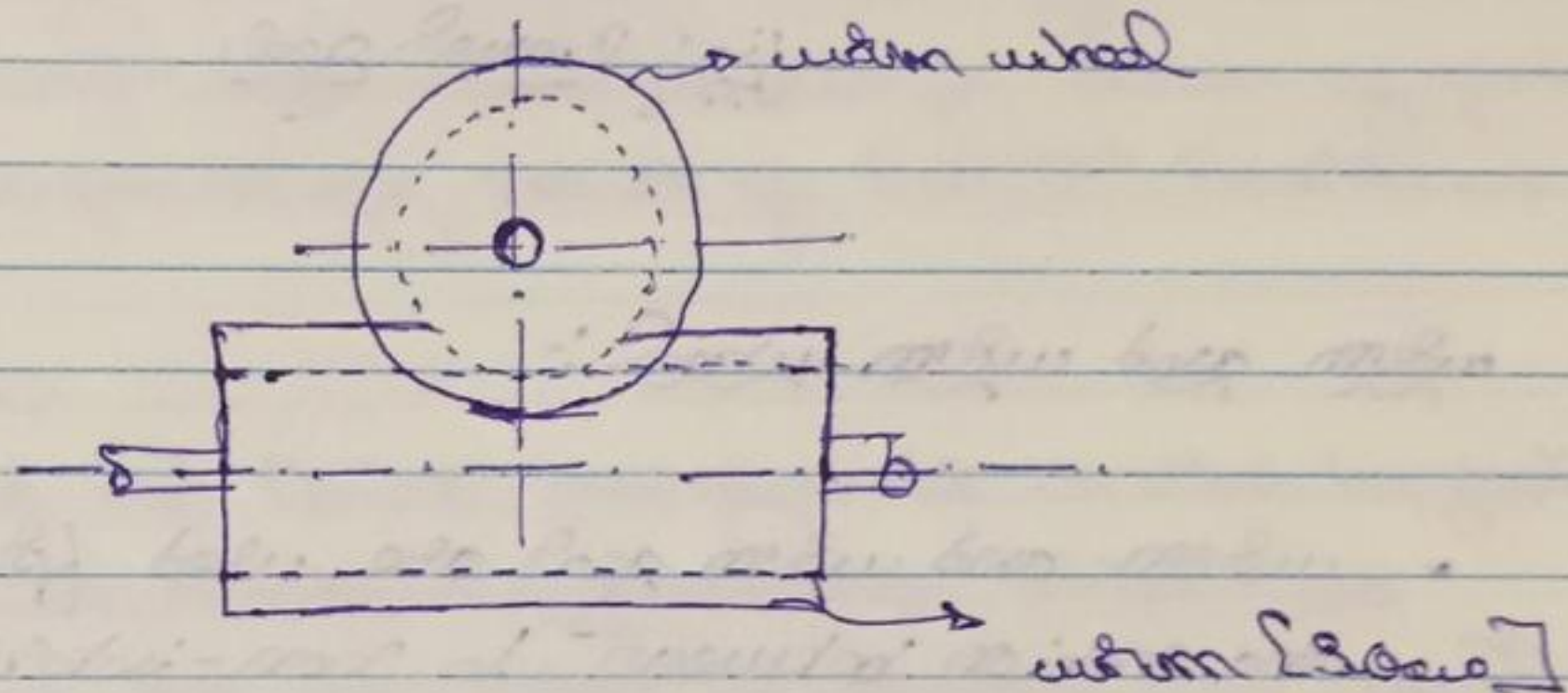
- worm gears are compact than the spur, helical and bevel gears.
- They can be used for high reduction ratios.
- These gears are self-locking (or) non-reversible.
- The operation of worm gears is silent & smooth.

Disadvantages:-

- The efficiency of worm gears is less.
- During the operation more heat is generated hence effective cooling is required.
- worm gears are more expensive than all gears.
- worm gears are difficult to manufacture.

Applications:-

- ~~worm~~ gears are commonly used in steering mechanisms, cranes, lawn mowers, lifts etc.
- They are also used in musical instruments like guitar, for locking the strings.



* Gear Trains:

- A gear train is a combination of two (or) more gears which is used to transmit motion from one shaft to another shaft.
- A gear train may consist of Spur, helical, bevel (or) Spiral gears.
- Gear train is necessary when it is required to obtain large speed reduction within a small space.
- Gear trains are commonly used in various machines [Lathe, milling, drilling etc], automobiles, clocks, ships, watches, etc.

* Types of Gear trains:

The gear trains are classified depending upon the arrangement of gears.

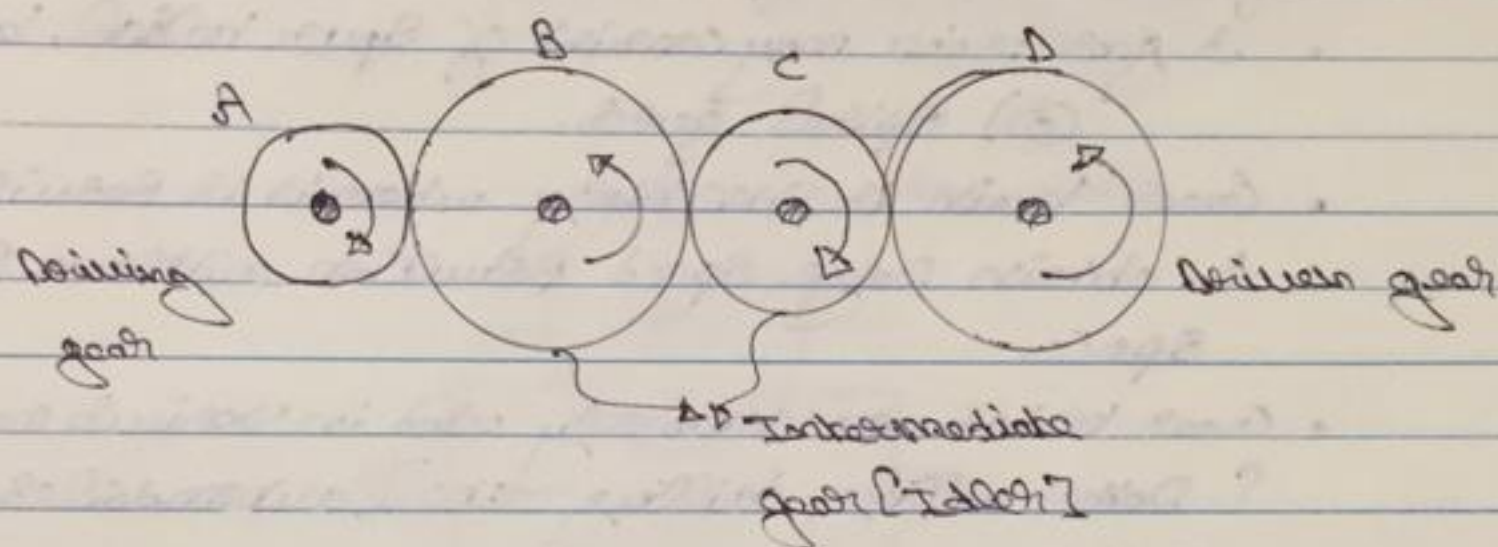
- ① Simple gear train.
- ② Compound gear train
- ③ Reverted gear train
- ④ Epicyclic gear train.

① Simple gear train

In simple gear train,

- Only one gear is mounted on each shaft.
- If gear A is the driver and gear D is the driven, then the motion is transmitted from A to D through the intermediate gears B & C, known as idler gears.
- If an odd number of idlers are used, the driver & the driven rotate in the same direction.

- If an even number (or) no idlers are used, the driver & the driven rotate in the opposite directions.



Velocity Ratio of Simple gear train: Let N_A, N_B, N_C, N_D be the speed and T_A, T_B, T_C, T_D be the number of teeth of the gears A, B, C, D respectively.

① A drives B

$$\frac{N_A}{N_B} = \frac{T_B}{T_A}$$

$$N_A T_A = N_B T_B$$

② B drives C

$$N_B T_B = N_C T_C$$

$$\frac{N_B}{N_C} = \frac{T_C}{T_B}$$

③ C drives D

$$N_C T_C = N_D T_D$$

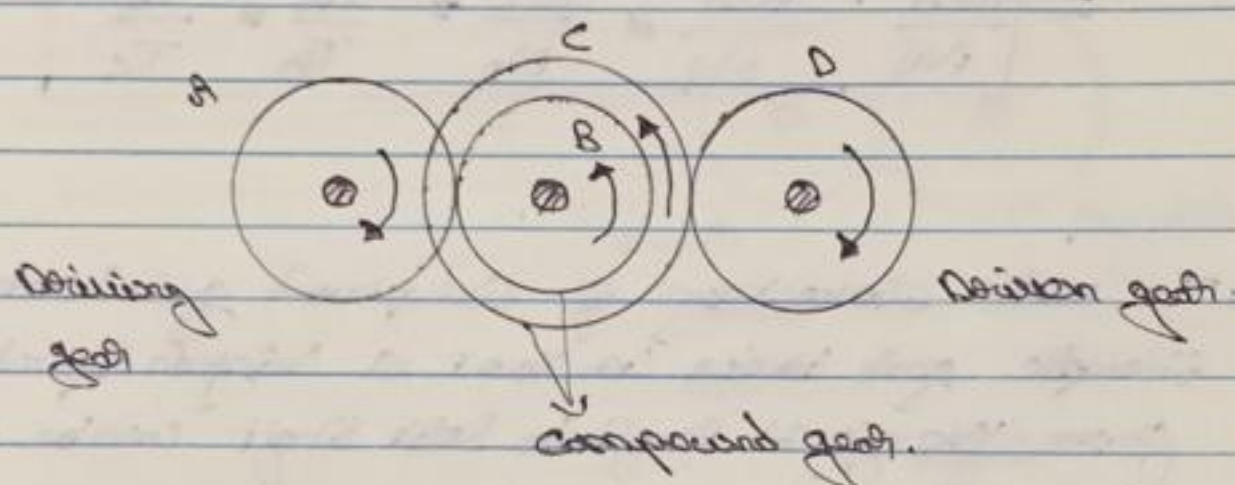
$$\frac{N_C}{N_D} = \frac{T_D}{T_C}$$

Velocity Ratio of Simple gear train = $\left\{ \frac{N_A}{N_D} \right\}$

$$\frac{N_A}{N_D} = \frac{N_A}{N_B} \times \frac{N_B}{N_C} \times \frac{N_C}{N_D} = \frac{T_B}{T_A} \times \frac{T_C}{T_B} \times \frac{T_D}{T_C} = \frac{T_D}{T_A}$$

* Compound gear Train

- If there are more than one gear on a shaft ~~except~~ except the first ~~gear~~ & last [driving and driven shaft] then it is called as compound gear train.



- In the above fig. A is driving gear, D is driven gear. B & C are mounted on intermediate shaft and are called compound gears.
- Since compound gears are mounted on the same shaft, they rotate at same speed & in the same direction.

* Velocity Ratio of compound gear train: Let N_A, N_B, N_C, N_D be the Speed and T_A, T_B, T_C, T_D be the number of teeth of the gears A, B, C, D respectively.

(i) A drives B

$$N_A T_A = T_B N_B$$

$$\frac{N_A}{N_B} = \frac{T_B}{T_A}$$

(ii) Since the gears B & C are mounted on the same ^{same} shaft, both of them rotate at the Speed.

$$N_B = N_C$$

$$\text{but } T_B \neq T_C$$

iii

Consider

$$N_c T_c = N_d T_d$$

$$\frac{N_c}{N_d} = \frac{T_d}{T_c}$$

∴ Velocity Ratio of compound gear train

$$= \frac{N_A}{N_D} = \frac{N_A}{N_B} \times \frac{N_c}{N_d} = \frac{T_B}{T_A} \times \frac{T_d}{T_c}$$

[*]. The main advantage of Compound gear train over Simple gear train is that, a higher Speed Reduction from the first shaft to last shaft can be obtained by using small gears.

- Hence, compound gear train requires less space as compared to Simple gear train for the given Speed Ratio.